



Workshop Materials
<https://tinyurl.com/4hdktc2w>

Beyond Arduino

Getting to “Blinky” (and beyond!) with a New
Microcontroller

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Schedule



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Intro

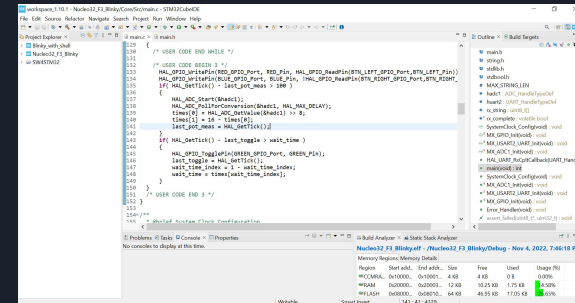
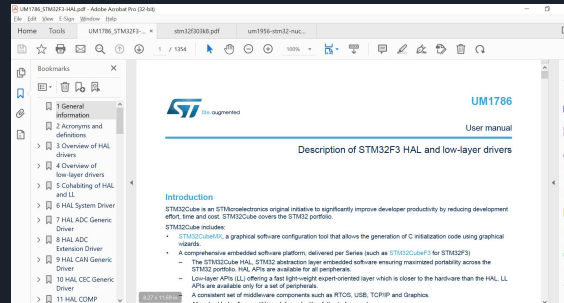
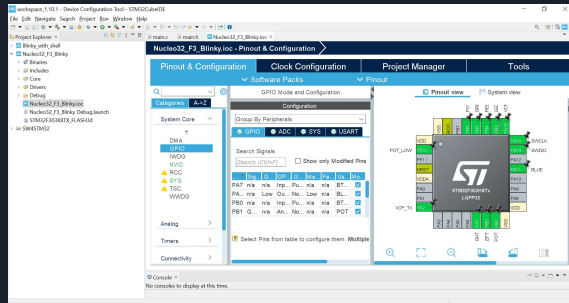
Scavenger Hunt

Digital out/UART Tx

Digital in / UART Rx

I2C

Projects / Closing





How to Do Hard Stuff



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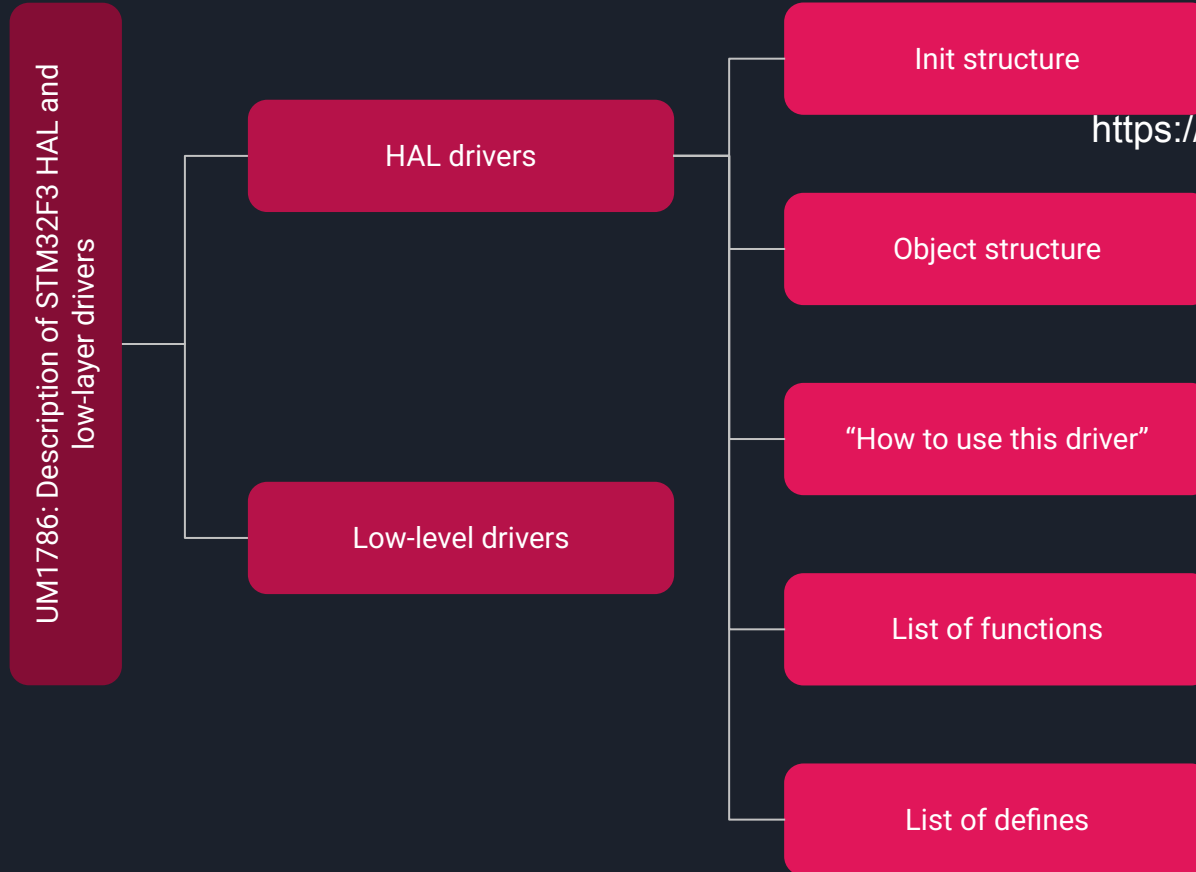
- Research
 - “Getting started” guide / User manual / Reference manual / Datasheet / App note
 - “___ appnote”
 - Google/forums (sometimes)
- Fail forward/Baby steps
- Debug
 - Change one thing at a time
 - Divide and conquer
 - Take breaks / Go for a walk outside
 - Write stuff down
 - Every now and then, start from scratch (e.g. cycle the power)
- Have fun!

How the STM32 HAL PDF is Organized



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First Project

Digital in/out & UART Tx/Rx

- Wire up a button & RGB LED
- Make sure to select pull-up/-down for the button inputs using the configuration list!
- Use:
 - HAL_GPIO_ReadPin
 - HAL_GPIO_WritePin
 - HAL_UART_Receive
- Check the workshop materials for project ideas!



Using a New Peripheral



Workshop Materials

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- Step #1: Use the default settings
 - #1.1 Build a basic application from scratch
 - Turn on the peripheral using either the configuration list or the pin mapping tool
 - Poke around the configuration list; which settings make sense and which don't? Click on the setting name for (some) more information.
 - Read the HAL, particularly the function list and "How to use this driver"
 - Start small:
 - Can you build without errors?
 - Can you send out data that you can also observe on a logic analyzer?
 - Can you send back data with a protocol decoder and read it?
 - Can you read the part or manufacturer IDs?
 - #1.2 Load and run a sample program
 - I2C_TwoBoards_ComPolling
 - I2C_TwoBoards_ComIT



Using a New Peripheral



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- Step #1: Use the default settings (continued)
 - #1.3 Recreate the sample program using the configuration tools.
 - #1.4 Modify the sample program or its configuration parameters.
 - Also look at:
 - Datasheet
 - Video series (Shawn Hymel, Phil's Lab, etc)
 - HAL source code
 - Books: Embedded Systems Fundamentals with Arm Cortex-M based Microcontrollers: A Practical Approach Nucleo-F091RC Edition (Alexander Dean)
- Step #2: When that doesn't work / When you need something other than the default settings
 - MCU reference manual
 - Appnotes (don't need to be from ST!)
 - Books!: Embedded Systems Fundamentals with Arm Cortex-M based Microcontrollers: A Practical Approach Nucleo-F091RC Edition (Alexander Dean)

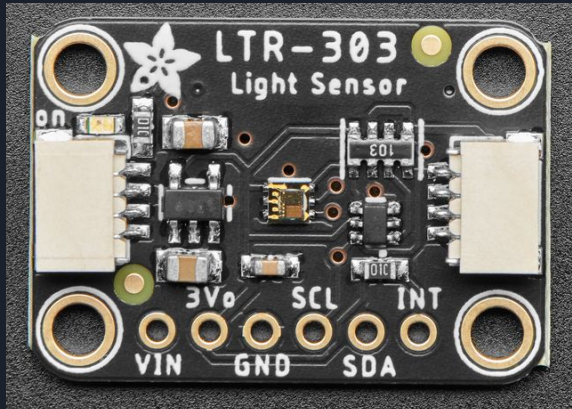
LTR-303 Light Sensor



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- Adafruit Part No. [5610](#)
- Device address is 0x29 (page 12 of the datasheet)
 - Must be left-shifted by 1 before using the HAL functions
- Read light data from register 0x88
 - MUST read 4 bytes, otherwise the LTR-303 won't take another measurement



Register Set				
Addr	R / W	Register Name	Description	Reset Value
0x80	R / W	ALS_CONTR	ALS operation mode control SW reset	0x00
0x85	R / W	ALS_MEAS_RATE	ALS measurement rate in active mode	0x01
0x86	R	PART_ID	Part Number ID and Revision ID	0xA0
0x87	R	MANUFAC_ID	Manufacturer ID	0x05
0x88	R	ALS_DATA_CH1_0	ALS measurement CH1 data, lower byte	0x00
0x89	R	ALS_DATA_CH1_1	ALS measurement CH1 data, upper byte	0x00
0x8A	R	ALS_DATA_CH0_0	ALS measurement CH0 data, lower byte	0x00
0x8B	R	ALS_DATA_CH0_1	ALS measurement CH0 data, upper byte	0x00
0x8C	R	ALS_STATUS	ALS new data status	0x00
0x8F	R/W	INTERRUPT	Interrupt settings	0x08
0x97	R/W	ALS_THRES_UP_0	ALS interrupt upper threshold, lower byte	0xFF
0x98	R/W	ALS_THRES_UP_1	ALS interrupt upper threshold, upper byte	0xFF
0x99	R/W	ALS_THRES_LOW_0	ALS interrupt lower threshold, lower byte	0x00
0x9A	R/W	ALS_THRES_LOW_1	ALS interrupt lower threshold, upper byte	0x00
0x9E	R/W	INTERRUPT_PERSIST	ALS Interrupt persist setting	0x00



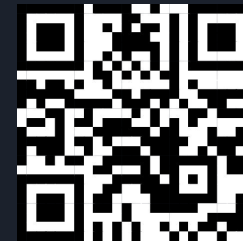
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Second Project

I2C + All previous

- Solder the header onto the light sensor, if you haven't already, wire it up, and configure the I2C peripheral on the STM32 (or don't!)
- Use:
 - HAL_I2C_Master_Transmit
 - HAL_I2C_Master_Receive
 - HAL_I2C_Master_Transmit_IT / HAL_I2C_MasterTxCallback
 - HAL_I2C_Master_Receive_IT / HAL_I2C_MasterRxCallback
- Check the workshop materials for project ideas!

Closing



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